

How the Proof Works

Take one vortex tube and stretch it along its axis. Incompressibility does not allow that extension to create volume, so the tube contracts across its width, narrowing the rotating core. Velocity changes across a shorter distance, the gradient steepens, and the vorticity becomes more concentrated even though the total kinetic energy can remain finite. This is the danger: the energy does not have to become infinite for the fluid to develop finer and more violent structure.

Navier–Stokes has a scaling that follows this concentration exactly. If (u, p) is a solution, then

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t)$$

preserves the balance between transport, pressure, and viscosity. Every term in the equation acquires the factor λ^3 . Meanwhile,

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Concentration can make a norm below $s = \frac{1}{2}$ smaller and a norm above it larger. The height $s = \frac{1}{2}$ does not move at all. It is the critical height at which zooming into the narrowing core leaves the size of the measurement unchanged.

That scaling gives the short Zeno calculation. Suppose one turn of the cascade contracts the spatial scale by a fixed factor $0 < \rho < 1$. A rescaled copy then has

$$r_j = \rho^j r_0, \quad U_j = \rho^{-j} U_0, \quad \tau_j = \rho^{2j} \tau_0.$$

The velocity scale rises as the core narrows, while the parabolic response time falls with the square of its width. Thus

$$\sum_{j=0}^{\infty} \tau_j = \tau_0 \sum_{j=0}^{\infty} \rho^{2j} = \frac{\tau_0}{1 - \rho^2} < \infty.$$

Scaling permits infinitely many finer turns to fit inside a finite interval. This calculation does not produce a singularity. It shows why finite energy and the parabolic scaling do not rule one out.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

What you'd expect to see in this pattern, if a singularity were to form, as you go up the time derivatives, you'd expect to see less spatial regularity, not more.

Now differentiate Navier–Stokes and watch what actually happens. Before any shorthand is introduced, the first three equations are

$$\partial_t u = \nu \Delta u - (u \cdot \nabla) u - \nabla p,$$

$$\partial_t^2 u = \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p,$$

$$\partial_t^3 u = \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p.$$

The nonlinear term opens once, then twice, and its multiplicities begin to appear. The complete pattern is the binomial form

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

Only now set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

The recurrence becomes

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n,$$

and pressure remains attached to every row through

$$-\Delta Q_n = \sum_{a=0}^n \binom{n}{a} \partial_i \partial_j ((V_a)_i (V_{n-a})_j).$$

Keep V_n in view. Its recurrence says that each move from V_n to V_{n+1} contains $\nu \Delta V_n$: the next temporal response exposes two spatial derivatives in the response immediately below it. The binomial transport terms and Q_n remain attached to that row throughout.

The critical height gives a compact readout of the spatial orders exposed by this continuing temporal climb. Let

$$\Lambda = (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

Each derivative $H^{(n)}$ is taken along the continuing $V_n = \partial_t^n u$ recurrence, where viscosity contributes $\nu \Delta V_n$, so successive temporal responses expose successive spatial energies:

$$H, H', H'', H''', \dots \quad \longmapsto \quad E_{1/2}, E_{3/2}, E_{5/2}, E_{7/2}, \dots$$

Higher temporal order opens higher spatial Sobolev order. As you do that, you get higher order Sobolev embeddings. Finite energy supplies the low frequencies; once the theorem carries every finite exposed height to one endpoint,

$$u \in \bigcap_{s < \infty} H^s(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The recurrence has exposed this direction, but an identity does not bound its terms. Fix a finite temporal depth N and a finite spatial height M . The datum-generated whole-terminal rectangle theorem gives

$$\sup_{0 < T < T_*} \mathcal{R}_{N,M}(u; T) < \infty,$$

with a constant allowed to depend on N and M . No common bound over all heights is being assumed. In particular, for every $0 \leq n \leq N$, the rectangle contains the adjacent pair

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_{n+1} \in L^2(0, T_*; H^{M-1}).$$

The adjacent rows give the temporal trace

$$V_n \in C([0, T_*]; H^M),$$

and the pressure equation supplies the corresponding Q_n -coordinates. This is a finite pressure-complete rectangle: finitely many temporal responses, finitely many spatial heights, their terminal traces, and the pressure required by the equation.

Now enlarge the rectangle. A larger choice of (N, M) does not construct new derivatives on the coordinates already present. Every rectangle is a restriction of the datum-generated maximal history, so two rectangles agree where they overlap. Their projective limit is

$$\mathcal{I}[u_0] := \varprojlim_{N, M < \infty} \mathcal{R}_{N, M}[u_0; T_*].$$

This is the all-orders intersection of the rectangles subject to agreement under every restriction, and its velocity coordinate satisfies

$$u \in \bigcap_{r, m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

Each finite coordinate was controlled before it entered this intersection, and compatibility makes all of those coordinates parts of one velocity–pressure history.

Take its zeroth temporal row. For every finite m , it has one trace in H^m , and agreement on the overlaps makes those traces one terminal velocity rather than a separate endpoint for each height:

$$u \in \bigcap_{m < \infty} C([0, T_*]; H_\sigma^m), \quad u_* := u(T_*) \in H_\sigma^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The recurrence and the pressure equation can now be evaluated at T_* , recovering every finite temporal and pressure coordinate of the terminal jet. Local well-posedness starts a smooth solution from u_* . On the interval where that restart meets the original history, uniqueness identifies the two, so the history continues past T_* . If T_* had been a finite maximal time, it was not maximal after all. Hence $T_* = \infty$.

The familiar finite estimates now come from selecting the coordinate they need. For finite temporal order r , spatial derivative order k , and $2 \leq p \leq \infty$, choose a finite m with

$$m > k + \frac{3}{2} - \frac{3}{p}.$$

On every finite time interval, the compatible intersection gives

$$\mathcal{I}[u_0] \longrightarrow H_t^1 H_x^m \longrightarrow C_t H_x^m \longrightarrow L_t^q W_x^{k, p}, \quad 1 \leq q \leq \infty,$$

for the chosen response $V_r = \partial_t^r u$. Different finite choices give, in immediate succession,

$$\begin{aligned} \mathcal{I}[u_0] &\longrightarrow C_t H_x^1 \longrightarrow L_t^\infty L_x^6, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^2 \longrightarrow L_t^\infty L_x^\infty, \\ \mathcal{I}[u_0] &\longrightarrow C_t H_x^3 \longrightarrow L_t^\infty W_x^{1, \infty}. \end{aligned}$$

No one of these estimates carries the argument. Any finite downstream demand selects a sufficiently high finite coordinate from the compatible intersection.

$$\begin{aligned} u_0 &\longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N, M}[u_0]\}_{N, M < \infty} \longrightarrow \mathcal{I}[u_0] \\ &\longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty. \end{aligned}$$